SECTION 7.B

Double-Angle and Pythagorean Identities

Learning Objectives

- I can apply double-angle identities for sine and cosine.
- I can rewrite double-angle identities in alternate equivalent forms.
- I can simplify trigonometric expressions using Pythagorean identities.
- I can justify each algebraic step when simplifying expressions.
- I can attend to precision when working in radians.

(Aligned to Learning Ladder Rung 3 — Module_07_Learning_Ladder)

Introduction

In Section 7A, we used structure in the unit circle to determine exact trig values.

Now we extend that structure.

Instead of evaluating trig functions directly from the unit circle, we will:

- Rewrite expressions
- Simplify complex trig forms
- Use identities to reveal hidden structure

Identities are equations that are always true for all values in their domain.

They allow us to:

- Simplify expressions
- Rewrite complicated forms
- Prepare expressions for solving equations
- Build proofs in later sections

Structure matters.

Key Vocabulary

Identity: An equation that is true for all values in its domain.

Double-Angle Identity: A formula that expresses a trigonometric function of 2θ in terms of θ.

Pythagorean Identities:

$\sin^2\theta + \cos^2\theta = 1$

$1 + \tan^2\theta = \sec^2\theta$

$1 + \cot^2\theta = \csc^2\theta$

Equivalent Forms: Different algebraic representations of the same trigonometric expression.

Core Identities

$\sin(2\theta) = 2\sin\theta\cos\theta$

$\cos(2\theta) = \cos^2\theta - \sin^2\theta$

$\cos(2\theta) = 2\cos^2\theta - 1$

$\cos(2\theta) = 1 - 2\sin^2\theta$

Structure Insight

Notice:

- The sine double-angle formula contains both sine and cosine.
- The cosine formula can be written three different ways.
- The alternate cosine forms come from substituting the Pythagorean identity.
- Recognizing which form to use is strategic.

Examples

Example 1: Evaluate $\sin(2\theta)$ if $\sin\theta = \frac{3}{5}$ and θ is in Quadrant I.

Example 2: Rewrite $\cos(2\theta)$ in terms of sine only.

Example 3: Simplify $\frac{1-\cos(2\theta)}{2}$.

Example 4: Simplify $\frac{\sin^2\theta}{1-\cos^2\theta}$.

Summary

- Identities are always true.
- Double-angle identities rewrite trig expressions involving 2θ.
- Cosine has three equivalent forms.
- Pythagorean identities allow substitution between sine and cosine.
- Justify each algebraic step.

Common Mistakes to Avoid

- Using the wrong double-angle identity.
- Forgetting to square sine or cosine.
- Stopping before fully simplifying.
- Treating identities as equations to solve instead of rewrite.

Practice Problems

Easy

1. $\sin(2\theta)$ if $\sin\theta = \frac{1}{2}$ and θ is in Quadrant I.

2. $\cos(2\theta)$ if $\cos\theta = \frac{4}{5}$ and θ is in Quadrant I.

3. Rewrite $\cos(2\theta)$ in terms of cosine only.

4. Simplify $\sin^2\theta + \cos^2\theta$.

Medium

5. Simplify $\frac{1-\cos(2\theta)}{2}$.

6. Rewrite $\cos(2\theta)$ in terms of sine only.

7. Simplify $\frac{\sin^2\theta}{1-\cos^2\theta}$.

8. Simplify $\frac{1+\tan^2\theta}{\sec^2\theta}$.

Harder

9. Simplify $\frac{\sin(2\theta)}{\cos\theta}$.

10. If $\sin\theta = -\frac{3}{5}$ and θ is in Quadrant III, find $\cos(2\theta)$.

11. Prove $\frac{1-\tan^2\theta}{1+\tan^2\theta} = \cos(2\theta)$.

12. Simplify $\frac{\sin^2\theta-\cos^2\theta}{\cos(2\theta)}$.

Challenge

13. Express $\cos(2\theta)$ entirely in terms of $\tan\theta$.

14. Simplify $\frac{2\sin\theta\cos\theta}{1-2\sin^2\theta}$.

15. If $\cos\theta = -\frac{8}{17}$ and θ is in Quadrant II, find $\sin(2\theta)$.

16. Verify $1 - 2\sin^2\theta = \cos(2\theta)$.

Justify

17. Explain why the three forms of $\cos(2\theta)$ are equivalent.

18. Justify why identities are always true and how that changes the way we simplify expressions.

Solutions

Easy

1. $\frac{\sqrt{3}}{2}$

2. $\frac{7}{25}$

3. $2\cos^2\theta - 1$

4. 1

Medium

5. $\sin^2\theta$

6. $1 - 2\sin^2\theta$

7. 1

8. 1

Harder

9. $2\sin\theta$

10. $\frac{7}{25}$

11. Identity verified.

12. -1

Challenge

13. $\frac{1-\tan^2\theta}{1+\tan^2\theta}$

14. $\tan(2\theta)$

15. $-\frac{240}{289}$

16. Identity verified.

Justify

17. Substitute Pythagorean identities into $\cos^2\theta - \sin^2\theta$.

18. Identities are true for all allowable values; we rewrite expressions rather than solve.